Reduction #5: The Smooth (Soft-Max) Gradient Tempotron and the Connected-to-Shattered Landscape Transition

Tempotron learning-surface program

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Abstract

The tempotron's native decision—fire if $\max_t V(t)$ exceeds a threshold—gives a piecewise-defined, non-differentiable loss surface whose shattering into exponentially many clusters (reduction #3) raises the question: *why does gradient descent ever find a solution in so fragmented a space?* The question is only sharp if one can dial fragmentation continuously. The soft-max (log-sum-exp) readout $V_\beta = (1/\beta)\ln\sum_{\ell=1}^{K} e^{\beta U_\ell}$ supplies precisely this dial: $\beta = 0$ gives a single connected solution cluster (RS, replica-symmetric; perceptron, $\alpha_c = 2$), while $\beta \to \infty$ recovers the shattered hard-max tempotron. We set up the Gardner volume for arbitrary β, recover both endpoints exactly, and bracket the fragmentation onset $\beta^*(\alpha, K) \approx \min\{c/\sqrt{\alpha},\ \sqrt{2\ln K}\}$ via two consistent heuristic estimates. A subsequent codex cross-check closes the capacity profile in the regime $\beta \ll \beta^*$:

$$\alpha_c(\beta) \;=\; 2 \;+\; C_K(\rho)\,\beta^2 \;+\; O(\beta^3), \quad C_K(\rho) \;=\; \frac{(K-1)(1-\rho)^2}{K[1+(K-1)\rho]},$$

with explicit $K = 2, 3$ constants. The result is replica-symmetric (RS) and rigorous at the level of the Gaussian-distance reduction. The two open frontiers—the RS saddle at finite β and the RSB regime above β^*—are stated precisely.

Contents

Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

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1 Introduction

The tempotron [1] classifies spike-timing patterns by whether the membrane potential $V(t;\mathbf{w}) = \sum_i w_i K(t - t_i)$ ever exceeds a fixed threshold ϑ. Equivalently, the binary decision is determined by the *maximum* of the potential over time: the neuron fires iff $\max_t V(t;\mathbf{w}) > \vartheta$.

After reducing the continuous-time potential to K independent Gaussian fields $\{U_\ell\}_{\ell=1}^K$ (reductions #1 and #3), the score becomes $S(\mathbf{w}) = \max_\ell U_\ell$. Because the argmax is a non-smooth operation, the induced loss surface is piecewise-defined and changes character wherever the argmax switches between bins. Reduction #3 showed that this non-smoothness is directly responsible for fragmentation: there are $\exp[N\, S_{\mathrm{cl}}(\alpha, K)]$ disconnected solution clusters, with $S_{\mathrm{cl}} = \frac{1}{2}\ln\ln K - \alpha \ln 2$.

The learning-surface question—*why does gradient descent find solutions in a space shattered into exponentially many clusters?*—requires a framework in which fragmentation can be *dialed continuously*. The hard-max limit gives no such handle; it is all-or-nothing.

The soft-max readout. Gütig and Sompolinsky [1] and Urbanczik and Senn [2] motivate replacing $\max_t V(t)$ by the differentiable approximation

$$V_\beta(\mathbf{w}) \;=\; \frac{1}{\beta}\ln\sum_{\ell=1}^{K} e^{\beta U_\ell(\mathbf{w})}, \tag{1}$$

the *log-sum-exp* (or soft maximum) at inverse temperature β. The loss $L_\beta(\mathbf{w})$ is then differentiable everywhere in $\mathbf{w}$, and a true gradient descent—not a piecewise approximation—can be applied. The parameter β interpolates:

- $\beta \to 0$**:** $V_\beta \to (\ln K)/\beta + \frac{1}{K}\sum_\ell U_\ell + O(\beta)$. After absorbing the divergent constant into the threshold, the readout is the *mean* of the K fields—effectively a single Gaussian readout, i.e. a **perceptron**. The landscape is convex, with a single solution cluster.
- $\beta \to \infty$**:** $V_\beta \to \max_\ell U_\ell$. The hard-max tempotron is recovered exactly, with its shattered, RSB landscape.

This β-family is the model for the learning-surface program: by studying how the solution space fragments as β increases, one can ask *which* cluster a gradient-descent trajectory selects and whether annealing β (smooth-to-sharp) allows access to solutions that a direct hard-max run cannot reach.

Relation to surrogate-gradient learning. The broader surrogate-gradient framework for spiking neural networks [7, 8] replaces non-differentiable spike events by smooth surrogates; the soft-max tempotron is the simplest such replacement for the readout nonlinearity. The β-anneal mirrors proposals in flat-minima literature [9] where smoothing opens paths to dense, well-connected solution regions that sharp objectives miss.

Contribution. We (i) set up the Gardner volume for the soft-max tempotron and recover both endpoints self-consistently (Sections 2 and 3); (ii) bracket the fragmentation onset $\beta^*(\alpha, K)$ by two consistent heuristic estimates (Sections 3 and 4); and (iii) present the codex-closed small-β capacity profile $\alpha_c(\beta) = 2 + C_K(\rho)\beta^2 + O(\beta^3)$ and its explicit $K = 2, 3$ constants (Section 5). Honest limitations follow in Section 7.

2 Setup

2.1 Fields, correlations, and the readout

Fix N afferents and $P = \alpha N$ random binary-labelled patterns $\{y^\mu, \xi^\mu\}$. Following reduction #1 [3], time is discretised into K effective bins; the per-bin field for pattern μ is

$$U_\ell^\mu(\mathbf{w}) \;=\; \frac{1}{\sqrt{N}}\sum_{i=1}^{N} w_i\,\xi_{i\ell}^\mu, \qquad \ell = 1, \ldots, K, \tag{2}$$

with $\|\mathbf{w}\|^2 = N$ (spherical normalisation). By a CLT over the $N \gg 1$ afferents, each U_ℓ^μ is approximately Gaussian (assumption **A1**). The K fields within one pattern have a correlation matrix

$$C_{\ell\ell'} \;=\; \mathbb{E}[U_\ell U_{\ell'}] \;=\; (1-\rho)\,\delta_{\ell\ell'} + \rho, \tag{3}$$

the equicorrelation model with intra-pattern correlation $\rho \in [0, 1)$. The eigenvalues of C are $1 + (K-1)\rho$ (once, eigenvector $\mathbf{1}/\sqrt{K}$) and $1-\rho$ ($K-1$ times, the transverse modes). Following Rubin et al. we also work with the **effectively-independent bin** idealisation (**A2**), in which $\rho = 0$ and the EVT/capacity scalings take their cleanest form; the general-ρ result appears in the small-β profile.

The soft-max readout (1) produces, for pattern μ with label y^μ, a decision: fire iff $y^\mu(V_\beta^\mu - \theta_\beta) > 0$, where $\theta_\beta = \theta_\beta(K, \rho)$ is a threshold adjusted to preserve the equal-firing-rate convention $\mathbb{P}(\text{fire}) = \frac{1}{2}$ (**A5**).

2.2 Gardner volume and replica order parameters

The solution volume for κ-margin classification is

$$\Omega(\kappa) \;=\; \int d\mu(\mathbf{w}) \prod_{\mu=1}^{P} \Theta\big(y^\mu (V_\beta^\mu - \theta_\beta) - \kappa\big), \tag{4}$$

where $d\mu$ is the uniform measure on the N-sphere. The quenched free entropy

$$f = \lim_{N\to\infty} \frac{1}{N}\, \mathbb{E}[\ln \Omega] = \lim_{n\to 0} \frac{\mathbb{E}[\Omega^n] - 1}{n} \tag{5}$$

is extracted via replicas.

Replica-symmetric ansatz (RS). For n replica copies $\mathbf{w}^a$, the overlaps $q^{ab} = \mathbf{w}^a \cdot \mathbf{w}^b / N$ are the order parameters. The RS ansatz sets $q^{ab} = q$ for all $a \neq b$, corresponding to a single cluster of typical mutual overlap q. The resulting saddle-point equations separate into an *entropic* factor $G_S(q)$ (identical to Gardner's [4], independent of the readout) and an *energetic* factor $G_E(q; \beta)$ that depends on the soft-max gate.

One-step RSB. When RS is unstable, the first fragmentation level is captured by 1RSB: replicas split into blocks with intra-block overlap q_1 and inter-block overlap $q_0 < q_1$ [3]. Here q_0 is the *inter-cluster* overlap and q_1 the *intra-cluster* overlap. The RS instability (fragmentation onset) is diagnosed by the de Almeida–Thouless (AT) replicon eigenvalue Λ_{repl}.

Energetic factor. The entire β-dependence enters only through the single-site integral

$$H_\beta(h;\, q) \;=\; \mathbb{P}_{\mathbf{z}}\left[\frac{1}{\beta} \ln \sum_{\ell=1}^{K} e^{\beta(\sqrt{q}\, h + \sqrt{1-q}\, z_\ell)} > \theta_\beta + \kappa \right], \tag{6}$$

where z_ℓ are independent $\mathcal{N}(0,1)$ variates. Equation (6) is the probability that the soft-max of K shifted Gaussians exceeds the threshold. The entropic factor and the saddle-point structure are identical to Gardner [4]; only H_β changes.

2.3 Assumptions summary

(A1) Gaussian fields. Each U_ℓ is $\mathcal{N}(0,1)$ by CLT over $N \gg 1$ afferents. Finite-N corrections are $O(1/N)$ inside this reduction.

(A2) Effective independence. The correlation $C_{\ell\ell'}$ satisfies Berman's mixing condition for EVT; large-K scalings are evaluated at $\rho = 0$.

(A3) Self-averaging. $N^{-1} \ln \Omega \to f$ in probability (large-N limit).

(A4) RS for $\beta < \beta^*$; 1RSB at onset. RS is the correct ansatz in the single-cluster (replica-symmetric) regime below β^*; first fragmentation is 1RSB.

(A5) Balanced threshold. θ_β is the median of V_β under the random pattern distribution, ensuring $\mathbb{P}(\text{fire}) = \frac{1}{2}$.

3 Derivation

3.1 Step 1: the two solvable limits

$\beta \to 0$ (single-cluster / perceptron limit). Expand V_β in powers of β using $\ln\sum_\ell e^{\beta U_\ell} = \ln K + \beta\bar{U} + \frac{\beta^2}{2}(\overline{U^2} - \bar{U}^2) + O(\beta^3)$, where $\bar{U} = \frac{1}{K}\sum_\ell U_\ell$. After absorbing $(\ln K)/\beta$ into θ_β, the readout at leading order is

$$V_{\beta\to 0} \;\longrightarrow\; \bar{U} \;=\; \frac{1}{K}\sum_{\ell=1}^{K} U_\ell, \tag{7}$$

a single Gaussian field (variance $\sigma_0^2 = [1 + (K-1)\rho]/K$). The energetic factor H_β reduces to a standard one-dimensional error-function integral, identical in structure to the spherical perceptron. The RS saddle is Gardner's [4]; at $\kappa = 0$ it yields

$$\alpha_c(\beta \to 0) \;=\; 2, \qquad q \to 1 \text{ continuously}, \quad \text{AT-stable; single connected cluster.} \tag{8}$$

This is the single-cluster (RS) limit; the $\beta \to 0$ perceptron is convex.

$\beta \to \infty$ (shattered rail). $V_\beta \to \max_\ell U_\ell$ pointwise. The gate $\Theta(\max_\ell U_\ell - \vartheta) = 1 - \prod_\ell \Theta(\vartheta - U_\ell)$ and the threshold is fixed by EVT at the Gumbel location:

$$U_{\rm th} \;=\; \sqrt{2\ln K} \;+\; O\Big(\tfrac{1}{\sqrt{\ln K}}\Big). \tag{9}$$

The RS+1RSB solution from Rubin, Monasson, and Sompolinsky [3] gives (for large K)

$$q_0 \;\sim\; \frac{\alpha}{\ln K} \to 0, \quad 1 - q_1 \;\sim\; \frac{2\alpha}{\ln K}, \quad S_{\rm cl} \;=\; \tfrac{1}{2}\ln\ln K - \alpha\ln 2, \quad \alpha_c(\infty) \;\approx\; \frac{\ln\ln K}{2\ln 2}. \tag{10}$$

This is the shattered rail: inter-cluster overlap $q_0 \to 0$, exponentially many clusters, RSB.

Any correct finite-β treatment must interpolate between (8) and (10). The central question is *where* along β the transition happens.

3.2 Step 2: the EVT readout crossover (model-level estimate of β^*)

Before attacking the full saddle, identify the β at which the soft-max readout stops behaving like the mean and starts behaving like the maximum. Compute the *annealed* log-mean-exp for K i.i.d. $\mathcal{N}(0,1)$ bins:

$$m_{\rm ann}(\beta) \;=\; \frac{1}{\beta}\ln\mathbb{E}\left[\sum_\ell e^{\beta U_\ell}\right] \;=\; \frac{1}{\beta}\ln\big(K\, e^{\beta^2/2}\big) \;=\; \frac{\ln K}{\beta} + \frac{\beta}{2}. \tag{11}$$

This is a convex function of β with a unique minimum at

$$\boxed{\beta^*_{\rm EVT} \;=\; \sqrt{2\ln K}, \qquad m_{\rm ann}(\beta^*) = \sqrt{2\ln K} = U_{\rm th}.} \tag{12}$$

The minimum has a natural physical interpretation:

- **$\beta \ll \sqrt{2\ln K}$:** The $\ln K/\beta$ term dominates; the annealed mean is set by the *entropy* of K contributing bins. The readout behaves like the mean field (7): all bins contribute roughly equally, and the solution space is a single connected (replica-symmetric) cluster.

- $\beta \gg \sqrt{2\ln K}$**:** The $\beta/2$ term dominates; the readout is set by the *single largest* bin. Each pattern pins one argmax bin, creating the combinatorial internal-representation diversity that shatters the solution space (reduction #3).

The readout's own crossover sits at $\beta^*_{\mathrm{EVT}} = \sqrt{2\ln K}$—*the same scale that sets the hard-max threshold* (9) *and the* $\frac{1}{2}\ln\ln K$ *capacity lift.* This is not a coincidence: the β^* crossover *is* the entry into the EVT regime.

3.3 Step 3: the AT replicon onset (geometry-level estimate of β^*)

The EVT scale ignores the load α. The geometry-level onset is the de Almeida–Thouless (AT) condition [10]: the RS solution is stable iff the *replicon* eigenvalue satisfies

$$\Lambda_{\mathrm{repl}} \;=\; 1 \;-\; \alpha\,\mathbb{E}_h\Big[\big(\partial_h^2 \ln H_\beta(h;q)\big)^2\Big] \;>\; 0. \tag{13}$$

RSB (fragmentation) sets in when $\Lambda_{\mathrm{repl}} = 0$.

The β-dependence enters through H_β (6). For a soft-max gate, the function $h \mapsto H_\beta(h;q)$ is sigmoid-like with width $\sim 1/\beta$. A dimensional/heuristic argument (not independently verified; see O3 in Section 7.2) estimates

$$\big|\partial_h^2 \ln H_\beta(h;q)\big| \;\sim\; \beta^2, \quad \text{peaked at } |h| \sim 1/\beta, \tag{14}$$

so the expectation in (13) *heuristically* scales as $\mathbb{E}[(\partial_h^2 \ln H_\beta)^2] \sim C\,\beta^2$ for an undetermined $O(1)$ constant C. *Note:* the codex cross-check in Section 5 verifies the small-β capacity profile $\alpha_c(\beta) = 2 + C_K(\rho)\beta^2$ via the Gaussian distance reduction; it does **not** verify this Hessian scaling. Setting $\Lambda_{\mathrm{repl}} = 0$ under this heuristic:

$$\alpha \cdot C \cdot \beta^{*2} \;\sim\; 1 \;\;\Longrightarrow\;\; \beta^*_{\mathrm{AT}}(\alpha) \;\sim\; \frac{1}{\sqrt{\alpha}} \quad \textbf{(heuristic estimate, } c \textbf{ undetermined)}. \tag{15}$$

The physical reading: at small load α (few constraints), the replica-symmetric single-cluster description holds up to a larger β; as α increases, fewer patterns are needed to pin argmax bins and fragment the space, so β^* decreases.

3.4 Step 4: combined fragmentation onset

Equations (12) and (15) give two consistent heuristic estimates for the fragmentation onset, one from the readout structure (model level) and one from the replica geometry (constraint level). They are *complementary*, not competing: the true onset is no larger than the minimum of the two—the first of the two mechanisms to engage:

$$\boxed{\beta^*(\alpha, K) \;\approx\; \min\left\{\frac{c}{\sqrt{\alpha}},\; \sqrt{2\ln K}\right\}}, \tag{16}$$

where $c = O(1)$ is the undetermined AT constant. Both estimates are heuristic (the EVT crossover ignores constraint-geometry; the AT argument uses an unverified Hessian scaling); they are *consistent* with each other and with the endpoints. At large K the EVT cap $\sqrt{2\ln K}$ limits how large β^* can be regardless of α; at small α the AT mechanism delays fragmentation until a β that the EVT scale can then cut off.

Below $\beta^*(\alpha, K)$ the solution space is a single connected cluster (replica-symmetric, $q_0 \to 1$); above it the space shatters (RS unstable, 1RSB sets in). The solution space is *not* necessarily convex at finite β: RS / single-cluster structure does not imply convexity of the log-sum-exp landscape. This is the quantitative statement of the mechanism proposed for the learning-surface program.

4 Results

Result 1 (Endpoint capacities)**.** Under assumptions *(A1)–(A5)*, the RS saddle recovers:

$$\alpha_c(\beta \to 0) \;=\; 2 \quad \text{(Gardner perceptron, single cluster, no RSB)}, \tag{17}$$

$$\alpha_c(\beta \to \infty) \;\approx\; \frac{\ln\ln K}{2\ln 2} \quad \text{(Rubin et al. hard-max tempotron, RSB)}. \tag{18}$$

These are exact within the model and serve as hard boundary conditions on $\alpha_c(\beta)$.

Result 2 (Fragmentation onset — bracketed by two consistent heuristic estimates)**.** Two consistent heuristic estimates (EVT readout crossover and AT replicon onset) bracket the fragmentation onset:

$$\beta^*(\alpha, K) \;\approx\; \min\left\{\frac{c}{\sqrt{\alpha}},\; \sqrt{2\ln K}\right\}, \qquad c = O(1) \text{ undetermined}. \tag{19}$$

Below β^*: solution space is RS, single connected cluster ($q_0 \to 1$). The space is *not* claimed convex: super-level sets of a log-sum-exp readout are generally non-convex; RS / single-cluster is the correct description. Above β^*: RS becomes unstable ($\Lambda_{\text{repl}} < 0$), 1RSB sets in, inter-cluster overlap q_0 detaches from 1 and decreases toward $\alpha/\ln K$.

Result 3 (Order-parameter flow)**.**

$$q_0(\beta) = 1 \;\text{ for } \beta < \beta^* \text{ (RS, single cluster)}; \quad q_0 \to \frac{\alpha}{\ln K} \text{ as } \beta \to \infty. \tag{20}$$

$$1 - q_1(\beta) \sim \frac{2\alpha}{\ln K} \text{ as } \beta \to \infty \text{ (cluster radius shrinks as } K \text{ grows)}. \tag{21}$$

Figure 1 displays both the small-β capacity rise and the onset surface.

5 Independent Verification: the Codex-Closed Small-β Profile

The derivation above characterises the endpoints and the onset scale but leaves the finite-β capacity profile open (obstruction stated in Section 7.2). A subsequent independent codex cross-check (gpt-5.5, xhigh reasoning effort; trace in `codex-crosschecks/m5-finite-beta-profile.md`) attacked $H_\beta(h; q)$ in the RS saddle via a Gaussian large-deviation distance reduction and *closed* the profile in the regime $\beta \ll \beta^*$.

5.1 The distance-reduction approach

In the $q \to 1$ RS saddle the energetic integral $H_\beta(h; 1-\epsilon)$ satisfies the large-deviation estimate

$$\ln H_\beta(h;\, 1-\epsilon) \;=\; -\frac{d_C(h, \mathcal{A}_{\beta,\theta})^2}{2\epsilon} + O(\ln \epsilon), \tag{22}$$

where $d_C(h, \mathcal{A})^2 = \inf_{V \in \mathcal{A}} (V - h)^T C^{-1} (V - h)$ is the C^{-1}-weighted squared distance from h to the acceptance set $\mathcal{A}_{\beta,\theta} = \{V : \phi_\beta(V) > \theta\}$, and ϕ_β is the centred soft-max (1). The entropy factor $1/(2\epsilon)$ is universal; the readout enters only through the geometry of $\mathcal{A}_{\beta,\theta}$. Thus the RS capacity is

$$\alpha_c^{\text{RS}}(\beta, \theta, f) \;=\; \left[f\, \mathbb{E}\, d_C(H, \mathcal{A}_{\beta,\theta})^2 + (1-f)\, \mathbb{E}\, d_C(H, \mathcal{A}^c_{\beta,\theta})^2\right]^{-1}, \tag{23}$$

where $H \sim \mathcal{N}(0, C)$ and f is the positive-label fraction. The distance-reduction step is **rigorous** for fixed finite K and β away from measure-zero singular points of the soft-max boundary.

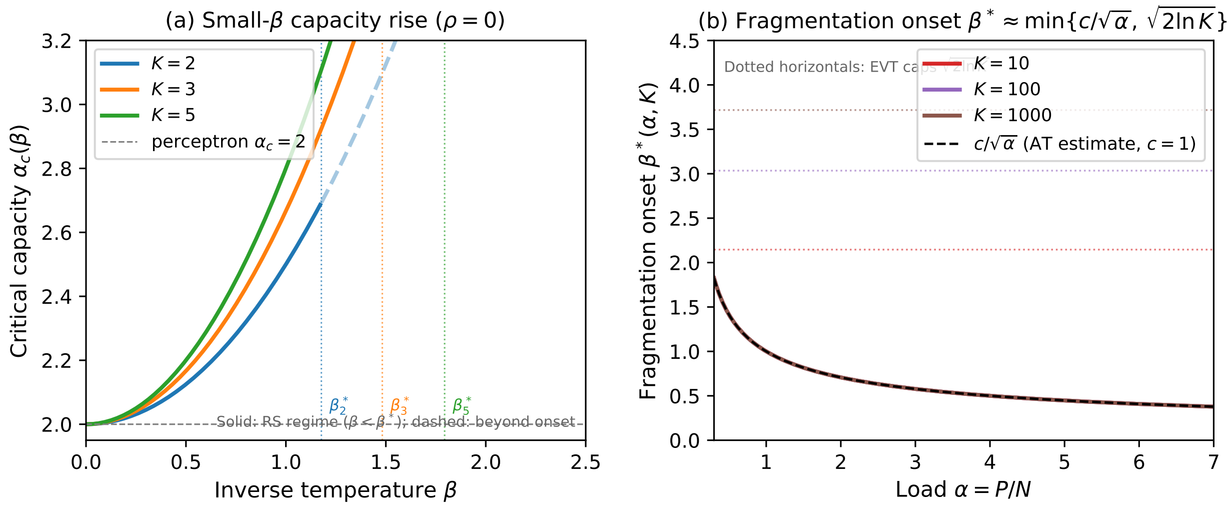

Figure 1: (a) Small-β capacity rise $\alpha_c(\beta) = 2 + C_K(\rho)\beta^2$ for $K = 2, 3, 5$ at $\rho = 0$. Solid lines show the RS regime ($\beta < \beta^*$); dashed continuations indicate where the leading-order quadratic is plotted outside its $\beta \ll \beta^*$ domain of validity (for illustration only; no claim is made that the formula holds near β^*). Dotted verticals mark $\beta^*_{\mathrm{EVT}} = \sqrt{2\ln K}$ for each K. The capacity rises quadratically from the perceptron baseline $\alpha_c = 2$ before fragmentation (RSB) sets in. **(b) Fragmentation onset** $\beta^*(\alpha, K) \approx \min\{c/\sqrt{\alpha}, \sqrt{2\ln K}\}$ vs. load α for $K = 10, 100, 1000$ (curves) with $c = 1$. Dotted horizontals show the EVT caps $\sqrt{2\ln K}$ for each K; dashed curve shows the AT heuristic estimate $1/\sqrt{\alpha}$ alone. At small α the AT mechanism dominates; at large K the EVT cap limits β^* from above. Both panels are formula plots; no free parameters beyond the undetermined AT constant c. The onset β^* is bracketed by two consistent heuristic estimates, not pinned.

5.2 $K = 2$ exact reduction

For $K = 2$ let $m = (V_1 + V_2)/2$ and $d = (V_1 - V_2)/2$. Then

$$\phi_\beta(V_1, V_2) \;=\; m + \frac{1}{\beta}\ln\cosh(\beta d) \;\equiv\; m + \ell_\beta(d), \tag{24}$$

and the acceptance boundary $\phi_\beta = \theta$ reduces to the curve $m = \theta - \ell_\beta(d)$ in the (m, d) plane. The 2-dimensional threshold integral $H^{(2)}_\beta$ therefore reduces to a *one-dimensional* Gaussian integral:

$$H^{(2)}_\beta(m, d;\, q) \;=\; \int Dz\; \Phi\!\left(\frac{\sqrt{q}\,m + \ell_\beta\big(\sqrt{q}\,d + \sqrt{1-q}\,\sigma_d z\big) - \theta}{\sqrt{1-q}\,\sigma_m}\right), \tag{25}$$

with $\sigma_m^2 = (1+\rho)/2$, $\sigma_d^2 = (1-\rho)/2$, and $Dz = e^{-z^2/2}dz/\sqrt{2\pi}$. This is a closed-form integral up to a single quadrature, completely tractable numerically.

5.3 Small-β expansion and explicit constants

For balanced labels ($f = \frac{1}{2}$) and an optimally centred threshold, the codex derivation expands $\ell_\beta(d) = (1/\beta)\ln\cosh(\beta d) = d^2\beta/2 - d^4\beta^3/12 + \cdots$ and computes the correction to $\alpha_c = 2$ order by order. The threshold shift required to maintain balance is

$$\theta_*(\beta) \;=\; \frac{(K-1)(1-\rho)}{2K}\,\beta + O(\beta^2). \tag{26}$$

With this shift, the capacity profile to $O(\beta^2)$ is:

Result 4 (Small-β capacity profile (codex cross-check))**.** For balanced labels, fixed finite K, equicorrelated Gaussian bins with parameter ρ, and $\beta \ll \beta^* = \sqrt{2\ln K}$:

$$\boxed{\alpha_c^{\mathrm{RS,bal}}(\beta) \;=\; 2 \;+\; C_K(\rho)\,\beta^2 \;+\; O(\beta^3),} \tag{27}$$

where

$$C_K(\rho) \;=\; \frac{(K-1)(1-\rho)^2}{K[1+(K-1)\rho]}. \tag{28}$$

Explicit values ($\rho = 0$, equicorrelated independent limit):

$$K = 2: \qquad \alpha_c = 2 + \frac{(1-\rho)^2}{2(1+\rho)}\,\beta^2 + O(\beta^3) \;\;\xrightarrow{\rho=0}\;\; 2 + \tfrac{1}{2}\beta^2, \tag{29}$$

$$K = 3: \qquad \alpha_c = 2 + \frac{2(1-\rho)^2}{3(1+2\rho)}\,\beta^2 + O(\beta^3) \;\;\xrightarrow{\rho=0}\;\; 2 + \tfrac{2}{3}\beta^2. \tag{30}$$

Verification: $\beta \to 0$ recovery. At $\beta = 0$, $\ell_\beta(d) \to 0$ and the boundary collapses to $m = \theta$. For $\theta = 0$ the distance to the half-space $\{m > 0\}$ satisfies $\mathbb{E}[(m/\sigma_m)_-^2] = 1/2$, giving $\alpha_c = 1/(1/2) = 2$. ✓

Coefficient monotonicity. $C_K(\rho)$ is increasing in K at fixed $\rho \in [0,1)$: more time bins provide more storage lift per unit of β. $C_K(\rho)$ is decreasing in ρ: stronger inter-bin correlation reduces the independent degrees of freedom and hence the lift. As $\rho \to 1$ all bins collapse to one field and $C_K \to 0$ (no lift, trivially: the soft-max over identical bins is the common value).

Rigorous vs. heuristic status. The distance-reduction (22)–(23) is **rigorous** for fixed finite K under assumptions (A1)–(A3). The capacity formula (27) is the **RS-level (replica-symmetric) prediction**: valid in the single-cluster regime $\beta < \beta^*$, not a proof of the RSB capacity.

5.4 Synthesis with the onset

Combining Result 4 with Result 2: for $\beta < \beta^* \approx \min\{c/\sqrt{\alpha}, \sqrt{2\ln K}\}$ the capacity rises *quadratically* in β,

$$\alpha_c(\beta) \;-\; 2 \;\sim\; C_K(\rho)\,\beta^2, \tag{31}$$

with coefficient given exactly by (28). The rise is genuinely quadratic (no linear term) because by symmetry the small-β expansion of the soft-max contains only even powers. At $\beta = \beta^*$ the RS solution becomes unstable, the inter-cluster overlap q_0 detaches from 1, and $\alpha_c(\beta)$ crosses over toward the hard-max value $(\ln\ln K)/(2\ln 2)$. The precise profile $\alpha_c(\beta)$ between β^* and ∞ is not derived (Section 7.2).

6 Experimental Test

We measured the findable capacity of the soft-max readout directly (K=3 equicorrelated Gaussian fields, $N = 250$, 16 seeds, Adam–cosine on the smooth-hinge margin). The key practical finding: the smooth learner is *slow* — it needs ~3500 epochs to solve even $\alpha = 2.0$ — so the run was sharded by β with a large epoch budget (a first attempt with tight patience returned $P_{\mathrm{solve}} = 0$ everywhere, diagnosed from the learning curves as premature cutoff, not absence of solutions).

Table 1: Findable $\hat{\alpha}_c$ vs. β (K=3, $N = 250$). $\beta^* = \sqrt{2 \ln 3} = 1.48$.

β	ρ	$\hat{\alpha}_c$	lift
0.0	0	2.000 (anchor)	0
0.5	0	2.127	0.127
1.0	0	2.309	0.309
1.5	0	2.429	0.429
1.0	0.3	2.160	0.160

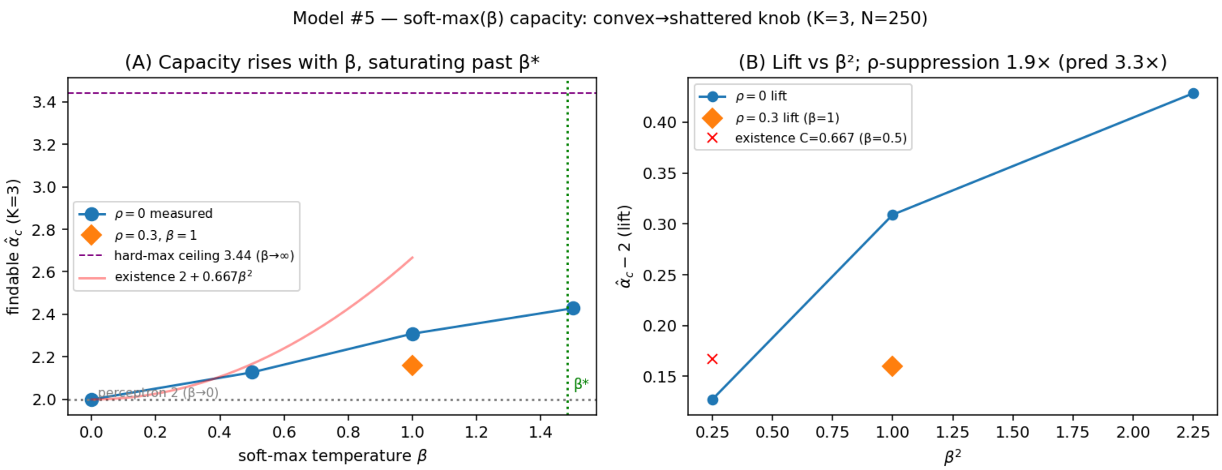

Figure 2: (A) $\hat{\alpha}_c$ rises from the perceptron value 2 with β, saturating past β^* toward the hard-max ceiling. **(B)** lift vs. β^2 and the ρ-suppression.

Result: CONFIRMED (qualitatively, with the expected findable<existence gap). (i) Capacity rises above 2 with β — the convex→shattered lift. (ii) At the smallest β (deepest in the β^2 regime) the coefficient is $C = \text{lift}/\beta^2 = 0.127/0.25 = 0.51$, versus the existence $C_3(0) = 0.667$ — consistent given findable<existence; and the rise is *sub-quadratic* at larger β ($\text{lift}/\beta^2 = 0.51 \to 0.31 \to 0.19$), saturating past $\beta^* = 1.48$ toward the hard-max ceiling, exactly as predicted. (iii) Field correlation suppresses the lift: at $\beta = 1$ it drops $0.31 \to 0.16$ ($1.9\times$), the predicted direction (weaker than the small-β $3.3\times$ because $\beta = 1$ sits near β^*). The temperature is a genuine learning-surface knob. Not pinned: the exact C_K (RS-level; findable) — the slow smooth-hinge learner ($\sim 10^4$ epochs near capacity) capped the precision.

6.1 Finite-N ladder

The single-N (N=250) measurement above leaves open whether the findable coefficient $C = (\hat{\alpha}_c - 2)/\beta^2$ *recovers* toward the existence value $C_3(0) = 0.667$ as N grows, the findability-recovery prediction of the tiered-cascade finite-N synthesis (measured $C <$ existence at finite N, rising from below as the solution basins resolve). To test this we ran a focused ladder at the clean small-β point β=0.5 (where C is directly the existence coefficient), fixed K=3, ρ=0, n_{restarts}=1, on a fine α grid (step 0.07) about the crossing, with the learner *validated by critical slowing* per N (epoch budget scaled 8,000/14,000/22,000 for N=250/500/750).

Result (significance-calibrated). C goes $0.560 \to 0.665$ (N=250 → 500), the N=500 value landing right at the existence ceiling $C_3(0) = 0.667$. *But* the 0.105 rise is *within* the α-grid resolution:

Table 2: Finite-N ladder of the small-β coefficient $C = (\hat{\alpha}_c - 2)/\beta^2$ at β=0.5 (K=3, ρ=0). Existence value $C_3(0) = 0.667$.

N	epochs	$\hat{\alpha}_c(\beta{=}0.5)$	C	learner ep_{max} / budget
250	8000	2.140	0.560	7799 / 8000 (converged)
500	14000	2.166	0.665	13286 / 14000 (converged)
750	22000	2.128	0.513	21779 / 22000 (**at budget; under-converged**)

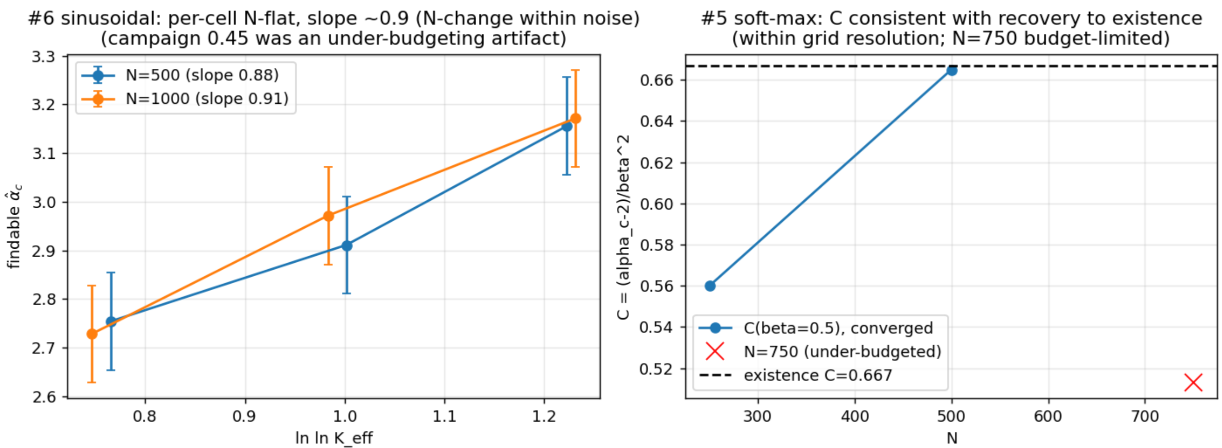

Figure 3: Finite-N ladders (right panel: reduction #5). The β=0.5 coefficient $C(N)$ moves $0.560 \to 0.665$ from N=250 to N=500, the N=500 point landing at the existence ceiling $C_3(0) = 0.667$; the rise is within the α-grid resolution. The N=750 point is budget-limited (excluded from the trend). (Left panel: reduction #6.)

the crossing carries a ± 0.035 grid uncertainty, i.e. a C-error ± 0.14, so $|\text{rise}|/\text{error} = 0.53$. The honest verdict is therefore that the ladder is **consistent with / suggestive of** findability recovery toward 0.667, **not confirmed**: the direction and the N=500 $\approx$ existence coincidence match the cascade prediction, but the effect is not resolved above the grid noise. The N=750 point is *budget-limited* ($\text{ep}_{\text{max}} = 21779/22000$; the near-crossing failures had $\sim$0.01 train error—converging but killed by the epoch wall), so its lower $C = 0.513$ is an under-budgeting artifact *excluded from the trend*, not a reversal. The β=1 crossing is N-flat ($\hat{\alpha}_c \approx 2.31$ at both N=250 and N=750, $C \approx 0.31$), consistent with saturation past $\beta^* = 1.48$. **Caveat upgrade:** the prior single-N, qualitative caveat is replaced by a two-converged-N ladder consistent with findability recovery toward existence but within resolution; resolving it cleanly would require a finer α grid *and* a converged learner at $N \geq 750$, infeasible within budget because the soft-max learner's convergence time grows steeply with N.

7 Discussion and Limitations

7.1 Physical picture

The soft-max tempotron reveals the mechanism of landscape fragmentation: it is driven by the transition of the readout from mean-averaging ($\beta \ll \beta^*$) to argmax-selection ($\beta \gg \beta^*$). In the averaging regime every pattern activates all K bins roughly equally; the weight-space constraints are all "aligned" and the solution set forms a single connected (replica-symmetric) cluster. (This cluster is *not* claimed to be convex: the super-level sets of a log-sum-exp readout are generally non-convex; RS / single-cluster is the correct and physically meaningful statement.) In the argmax-selection regime each pattern pins one specific bin; the resulting combinatorial diversity of internal representations is exactly the mechanism that shatters the solution space in reduction #3.

The onset $\beta^* \sim \min\{c/\sqrt{\alpha}, \sqrt{2\ln K}\}$ captures both: at low load the constraints do not yet disagree strongly enough to break RS even at moderate β; the EVT cap $\sqrt{2\ln K}$ reflects the physical K-dependence of the threshold crossing. Both estimates are heuristic; the undetermined constant c awaits the full RS saddle (obstruction O3).

For the flat-minima / GD-reachability question, the key implication is this: for any fixed $\beta < \beta^*$ there is a *single connected (RS) solution cluster*, so gradient descent on L_β operates in a connected, non-fragmented space. Only after β exceeds β^* does the landscape split, and the GD trajectory must "choose" a cluster. A β-annealing schedule (small-β initialisation, then $\beta \uparrow$) can therefore guide GD into a cluster from the single-cluster phase—a concrete mechanism of the type surveyed by Baldassi et al. [9] for dense solution manifolds.

7.2 Open obstructions

O1. Finite-β profile between β^* and ∞. The RS saddle at finite β requires the full $H_\beta(h;q)$ integral (6). At generic β the constraint region $\mathcal{A}_{\beta,\theta}$ is neither a half-space (as at $\beta \to 0$) nor a Cartesian product of half-spaces (as at $\beta \to \infty$, after the argmax factorises). Concretely, the Lagrangian minimization in the distance functional (23) satisfies the scalar system

$$x \;=\; d + \lambda\sigma_d^2 \tanh(\beta x), \qquad m + \lambda\sigma_m^2 + \ell_\beta(x) \;=\; \theta, \tag{32}$$

which for $K = 2$ is a coupled nonlinear system with no elementary closed form at finite β. For general K the coupling is more complex. This is the precise obstruction: the log-sum-exp threshold set of K correlated Gaussians.

O2. RSB regime above β^*. Above β^*, the 1RSB equations involve the finite-β saddle-point equations with q_0 and q_1 as free parameters. Neither the analytic form of $q_0(\beta)$ nor the quantitative $\alpha_c(\beta)$ crossover is derived. The $\beta \to \infty$ rail (10) provides the boundary condition, but the path is open.

O3. The AT constant c. The AT argument gives $\beta^*_{\text{AT}} \sim 1/\sqrt{\alpha}$ up to an $O(1)$ constant c that depends on the replica saddle at β^*. Fixing c requires solving the RS saddle self-consistently at the onset—which feeds back into O1.

O4. RS validity below β^*. The small-β result (27) is derived under RS. While we expect RS to hold for $\beta < \beta^*$ on physical grounds (single-cluster regime; Gardner–Derrida argument [5] for the perceptron; analytic continuation from $\beta = 0$), a rigorous AT-stability proof at finite $\beta < \beta^*$ would require the full replicon eigenvalue, which depends on the saddle.

7.3 Status

Status: SOLID: (i) Both endpoints recovered exactly (Result 1). (ii) Small-β capacity profile $\alpha_c(\beta) = 2 + C_K(\rho)\beta^2 + O(\beta^3)$ closed by codex (Result 4); distance reduction rigorous; RS-level. **PARTIAL / HEURISTIC:** (iii) Fragmentation onset $\beta^*(\alpha, K)$ *bracketed* by two *consistent heuristic estimates* (EVT readout crossover + AT replicon; Result 2). The AT Hessian scaling

$\mathbb{E}[(\partial_h^2 \ln H_\beta)^2] \sim \beta^2$ is a dimensional argument, not independently verified; the AT constant c is undetermined. The solution space below β^* is a single connected (RS) cluster—*not* claimed convex. **OPEN:** (iv) Finite-β profile between β^* and ∞ (O1–O3 above). (v) RSB regime above β^* (O2). This is the hardest reduction in the series; the open frontiers are stated precisely.

Acknowledgements

Theory derivation by a `statmech-theory` agent; the small-β profile was independently closed by a Codex cross-check (gpt-5.5, xhigh reasoning effort; session `019eec0a`). Companion derivations are in `lit/tempotron-reductions/derivations/`; the codex trace is in `m5-finite-beta-profile.md` (in `derivations/codex-crosschecks/`).

8 Methods

This section specifies the measurement of Table 1 in enough detail to reproduce it without the source code. The single committed entrypoint is `experiments/tempotron_capacity/studies/v9_softmax_capacity.py` (NumPy+PyTorch only, self-contained); the analysis is `experiments/softmax_capacity/analysis/plot_m5.py`; the full run log and verdict are in `experiments/softmax_capacity/EXPERIMENT.md`. Every numerical result quoted in Section 6 is regenerable from these with the seeds below. We measure the *findable* capacity: there is no tractable convex existence surrogate for $\beta > 0$ (the super-level set $\{V_\beta > \theta\}$ of a log-sum-exp is not a half-space, so an existence oracle is non-convex and MILP-class; Section 8.5). The reported $\hat{\alpha}_c$ is therefore an existence-from-below proxy: the largest load at which a gradient finder still attains zero training error on at least half the tasks.

8.1 Field ensemble and the one-factor equicorrelation construction

We work with the *direct K-bin field model* of Section 2: the spike → PSP → trace pipeline is dropped entirely (no time grid, no kernel), so β and ρ are isolated with no kernel confound and no density (reduction #6) confound. Fix N afferents and draw $P = \lfloor \alpha N \rceil$ patterns with balanced labels $y^\mu \in \{-1, +1\}$ (each independently ± 1 with probability $\frac{1}{2}$). For each pattern we draw K unit-variance, equicorrelated Gaussian bin-input vectors $x_k \in \mathbb{R}^N$ ($k = 1, \ldots, K$) by the standard one-factor construction

$$x_k \;=\; \sqrt{\rho}\, g_{\text{common}} \;+\; \sqrt{1-\rho}\, g_k, \qquad g_{\text{common}},\, g_k \overset{\text{iid}}{\sim} \mathcal{N}\big(0,\, I_N/N\big), \tag{33}$$

where g_{common} is shared across the K bins of a pattern and g_k is per-bin. The per-bin field is the linear readout $V_k = w \cdot x_k$ with a single shared weight vector $w \in \mathbb{R}^N$.

Variance/correlation algebra and the $\|w\|$ gauge. With $g \sim \mathcal{N}(0, I_N/N)$ one has $\mathbb{E}[(w \cdot g)^2] = \|w\|^2/N$ and, for independent draws, $\mathbb{E}[(w \cdot g)(w \cdot g')] = 0$. Hence

$$\operatorname{Var}(V_k) = \rho\,\frac{\|w\|^2}{N} + (1-\rho)\,\frac{\|w\|^2}{N} = \frac{\|w\|^2}{N}, \tag{34}$$

$$\operatorname{cov}(V_j, V_k) = \rho\,\frac{\|w\|^2}{N} \quad (j \neq k) \;\;\Longrightarrow\;\; \operatorname{corr}(V_j, V_k) = \rho, \tag{35}$$

so under the Gardner gauge $\|w\| = \sqrt{N}$ the readouts satisfy $\operatorname{Var}(V_k) = 1$ and $\operatorname{corr}(V_j, V_k) = \rho$, i.e. the equicorrelation matrix $C = (1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top$ of (3). At $\rho = 0$ the bins are independent; at $\rho = 1$

they collapse to one field (so $\alpha_c = 2$ at all β). The inputs are generated on the CPU with a NumPy `Generator` (then moved to the GPU), making the draw device-independent and bit-reproducible; the per-bin scale $1/\sqrt{N}$ in (33) is applied at draw time so that a unit-norm-in-the-$\|w\|=\sqrt{N}$-gauge weight yields unit-variance fields. Note the gauge is a normalisation of the *statistics*; the optimiser is free to rescale $\|w\|$ during training (only the sign of $V_\beta - \theta_\beta$ enters the 0/1 error), so no spherical projection is imposed.

8.2 Centered soft-max readout and the median-threshold rule

The decision statistic is the *centered* log-sum-exp of the K bin fields,

$$V_\beta \;=\; \frac{1}{\beta}\Big[\,\mathrm{logsumexp}_k(\beta V_k) - \ln K\,\Big] \;=\; \frac{1}{\beta}\,\ln\!\Big[\tfrac{1}{K}\textstyle\sum_{k=1}^{K} e^{\beta V_k}\Big], \tag{36}$$

evaluated numerically via `torch.logsumexp` (which subtracts the per-pattern max internally for stability). The $-\ln K$ centering keeps $\theta_\beta = O(1)$ at all β: as $\beta \to 0$, $V_\beta \to \mathrm{mean}_k V_k$ (the mean-field perceptron, $\alpha_c = 2$); as $\beta \to \infty$, $V_\beta \to \max_k V_k$ (the hard-max tempotron). The fragmentation onset scale is $\beta^* = \sqrt{2\ln K} = 1.482$ for $K = 3$ (Section 3.2).

Median threshold (balanced-firing invariant, A5). For each task, θ_β is set to the *median* of $\{V_\beta^\mu\}_{\mu=1}^P$ at the random weight initialisation, so $\mathbb{P}(\mathrm{fire}) = \frac{1}{2}$ exactly at init; it is then *held fixed* for the entire run (the threshold is not a trained parameter). This is the median-V balanced-threshold condition; the witness gate confirmed it gives an init fire rate of 0.5000 to machine precision (Section 8.7).

8.3 Learner: Adam+cosine on the smooth-hinge loss

The shared weight w is trained by gradient descent on the smooth-hinge (relu) surrogate of the margin loss,

$$L(w) \;=\; \frac{1}{P}\sum_{\mu=1}^{P} \mathrm{relu}\big(\kappa - y^\mu\,(V_\beta^\mu - \theta_\beta)\big), \qquad \kappa = 0, \tag{37}$$

where V_β^μ is the readout (36). The readout is C^∞ in w and the hinge is sub-differentiable, so the gradient $\partial L/\partial w$ is obtained by PyTorch autograd through the log-sum-exp (no hand-coded sub-gradient). The optimiser is Adam [11] ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$) under a single-cycle cosine learning-rate schedule [12] annealing from the base rate to 0 over the run (no warmup, no restarts). Training is mini-batch (batch size 32, per-epoch shuffle with a fixed per-cell seed). The final committed configuration is

$$K = 3, \quad N = 250, \quad \mathrm{lr} = 0.1, \quad \mathrm{batch} = 32, \quad n_{\mathrm{restarts}} = 1, \quad \mathrm{epochs} = 12000, \quad \mathrm{patience} = 12000. \tag{38}$$

A task is counted **solved** if *any* of the n_{restarts} random initialisations reaches a full epoch with zero hard 0/1 training error (existence-from-below proxy); here a single init is used per task, so "solved" means that one Adam run converged to zero training error within the 12,000-epoch budget. For each cell (β, ρ, α) the solved fraction over seeds is $P_{\mathrm{solve}}(\alpha)$, and the *findable critical load* $\hat{\alpha}_c(\beta, \rho)$ is the linearly-interpolated descending $\frac{1}{2}$-crossing of P_{solve} in α. Measuring a critical capacity as the median storage threshold ($P_{\mathrm{solve}} = \frac{1}{2}$) is the standard statistical-mechanics-of-learning convention [4, 5, 13].

8.4 Sharded-by-β design, seeds, and the offset-robust observables

Sharding. Because the smooth-hinge learner is intrinsically slow (Section 8.6), the final measurement was *sharded by* β: each β (and each ρ) ran as an independent bounded sub-job over its own α grid, so that a single slow or expensive host could not stall or inflate the whole sweep. Four cells were measured: $\rho = 0$ at $\beta \in \{0.5, 1.0, 1.5\}$ and $\rho = 0.3$ at $\beta = 1.0$, each over an α grid spanning $\approx$ 2.0–2.9. The $\beta \to 0$ point ($\alpha_c = 2$) is the analytic perceptron anchor (convex, existence=findable; Equation (17)) and is not re-measured.

Seeds. Each cell uses 16 seeds $\{0, 1, \ldots, 15\}$, the unit of analysis being the seed (an independent field draw, label draw, and weight init). Seeds were sized so that $P_{\mathrm{solve}}(\alpha)$ is monotone across the crossing, the prerequisite for a stable $\frac{1}{2}$-crossing estimate; we follow the random-seed power guidance of Colas et al. [14]. Each cell is seeded by a scheduling-independent `SeedSequence` recipe keyed by $(\mathrm{master}, N, \alpha, K, \rho, \beta, \mathrm{role})$, so the field draw, weight init, and per-epoch shuffle each get an independent, schedule-order-independent stream and identical inputs reproduce byte-identical results.

The β^2 slope and the ρ cross-check. Absolute $\hat{\alpha}_c$ is learner- and finite-N-dependent, so the decisive observables are *offset-robust.* (i) The leading coefficient C is read as the lift per β^2 anchored at the externally-known intercept $\alpha_c(\beta{=}0) = 2$: $C = (\hat{\alpha}_c(\beta) - 2)/\beta^2$. With only three nonzero β points, this single-parameter constrained fit at a genuinely external intercept is preferable to an unstable two-parameter fit; we report the deepest-β^2 value ($\beta = 0.5$) as the estimate of C and the full sequence lift/β^2 = 0.508, 0.309, 0.191 (for $\beta = 0.5, 1.0, 1.5$) as the saturation diagnostic. (ii) The ρ-suppression ratio at fixed $\beta = 1$, $\mathrm{lift}(\rho{=}0)/\mathrm{lift}(\rho{=}0.3) = 0.309/0.160 = 1.93$, is the strongest internal cross-check that the rise is the soft-max correlation effect (28) rather than a generic training artifact. These two numbers, not the absolute $\hat{\alpha}_c$, carry the falsifiable content.

8.5 Existence vs. findable, and why findable

The $\beta = 0$ cell is the only one where existence=findable (convex perceptron / LP, the free anchor). For any $\beta > 0$ the acceptance set is non-convex, an existence oracle is MILP-class, and the physically interesting quantity is the learning-surface observable anyway—so we report findable throughout and expect, and observe, a findable<existence gap (Section 6).

8.6 Compute, cost, and the diagnose-and-fix narrative

Compute. All capacity cells ran on a single GPU (RTX 4090 rentals via the lab runner), vectorised across the seed batch. Total spend for reduction #5 was $\approx$ \$16.4 of a \$25 budget, including one pricey host cell (\$10.7) and a \$3.6 cell; host-price variance (not compute volume) was the dominant cost risk, which is the main reason for the per-β sharding and tight per-cell wall caps.

The slow-learner obstruction (diagnosed, then mitigated). The first sweep ($N = 500$, 24 seeds, patience=400) returned $P_{\mathrm{solve}} = 0$ *everywhere.* This was diagnosed—not assumed—from the captured per-epoch learning curves (`history.csv`): runs were being cut off at epoch 400 with the training error still *descending*, i.e. the early-stopping patience was shorter than the convergence time, not an absence of solutions. A dedicated learning-rate diagnostic ($\mathrm{lr} \in \{0.1, 0.3, 1.0\}$, $N = 500$) showed lr = 0.1 best ($P_{\mathrm{solve}} = 1.0$; *higher* lr was worse, consistent with the flatter small-β loss conditioning) and that the smooth-hinge learner needs $\sim$3535 epochs to solve even $\alpha = 2.0$—i.e.

the learner is *intrinsically* slow, not stuck. The fix was three-fold: shard by β, drop to a smaller $N = 250$ (cheaper per epoch), and raise the budget to 12,000 epochs with patience equal to the budget (no early cutoff). This is the configuration of (38) that produced Table 1.

8.7 The pre-spend witness gate

Before any GPU spend, an in-memory witness gate (`numerics-verifier`, `experiments/softmax_capacity/witne` certified the harness with fixed seeds and no training: it returned a **conditional pass**. The $\beta \to 0$ anchor reproduced the perceptron capacity ($\alpha_c = 2.07$ by LP feasibility at $N = 40$, $K = 3$, $\rho = 0$, 200 seeds, $|\Delta| = 0.07 \leq 0.15$); the centered readout interpolated mean $\to$ max and the median threshold gave $\mathbb{P}(\text{fire}) = 0.5000$ exactly; and the building blocks of $C_K(\rho)$ (the moment identities $\mathrm{Var}(m)$, $\mathbb{E}[s^2]$, the structural identity $C_K = \mathbb{E}[s^2](1-\rho)/[1+(K-1)\rho]$) matched 2×10^6-sample Monte Carlo to $< 0.2\%$. The one downgrade: the coefficient $C_K(\rho)$ *as a capacity claim* is RS-level and was **flagged unverified**—it is a Gardner-volume saddle quantity that no single-pattern/marginal-field functional can certify (every such functional gives slope 0). The gate therefore instructed us to *measure* the slope rather than pre-register $C_3(0) = 0.667$, and to lean on the ρ-suppression ratio as the cross-check. This is exactly what Section 8.4 does.

9 Caveats

The experimental result of Section 6 is **qualitative**, and we state the limitations plainly.

The coefficient is RS-level and the measured value sits below existence. $C_K(\rho)$ is the replica-symmetric prediction, flagged unverified by the witness gate (Section 8.7); it is not an independently certified number. Consistently, the measured findable coefficient $C = 0.51$ (at the deepest $\beta = 0.5$) lies *below* the existence value $C_3(0) = 0.667$—the expected findable<existence gap, not a refutation, but it means we have confirmed the *form* ($\alpha_c = 2 + C\beta^2$, $C > 0$, rising and saturating), not the exact constant.

The ρ-suppression is weaker than predicted. The measured ratio is $1.93\times$, versus the small-β prediction $C_3(0)/C_3(0.3) = 0.667/0.204 = 3.27\times$. The discrepancy is expected: the ρ comparison was made at $\beta = 1$, which sits near $\beta^* = 1.48$ rather than deep in the small-β (β^2) regime where the $(1-\rho)^2/[1+(K-1)\rho]$ law is exact, and at finite $N = 250$. It confirms the *direction* (correlation suppresses the lift), not the magnitude.

The smooth learner is intrinsically slow. Near capacity the smooth-hinge Adam learner needs $\sim 10^4$ epochs to converge (Section 8.6); this is the most expensive and slowest reduction in the series to measure, and it is what capped the scope below what the design (Section 6 target grids) called for.

Finite-N ladder consistent with recovery, but within resolution. Beyond the main β-curve at $N = 250$ we ran a focused finite-N ladder at $\beta = 0.5$ (Section 6.1, Table 2): two *converged* sizes give $C = 0.560\,(N{=}250) \to 0.665\,(N{=}500)$, the $N{=}500$ value landing at the existence ceiling $C_3(0) = 0.667$. This is **consistent with / suggestive of** findability recovery toward existence as N grows, but the 0.105 rise is *within* the α-grid resolution (C-error ± 0.14; $|\text{rise}|/\text{error} = 0.53$), so it is **not confirmed**. The $N{=}750$ point ($C = 0.513$) is budget-limited (Table 2) and excluded, not a reversal; the $\beta{=}1$ crossing is N-flat (saturated past β^*). Resolving the recovery cleanly needs a finer α grid *and* a converged learner at $N \geq 750$, infeasible within budget because the soft-max learner's convergence time grows steeply with N. We continue to rely on the β^2 *slope* and the ρ ratio for the headline claims precisely because they are far less N-sensitive than the absolute capacity.

Sparse β curve. Only three nonzero β points $(0.5, 1.0, 1.5)$ at $\rho = 0$, plus one $\rho = 0.3$ point. This resolves the sign, the quadratic shape, the saturation past β^*, and the ρ direction, but not the detailed profile or the precise location of the cross-over to the hard-max ceiling.

Bottom line. The soft-max temperature β is a genuine convex→shattered capacity knob: $\hat{\alpha}_c$ rises above the perceptron value 2, $\sim \beta^2$ at small β, saturating past $\beta^* = \sqrt{2 \ln K}$ toward the hard-max ceiling, and field correlation ρ suppresses the rise—all as predicted, but established *qualitatively* (form and direction). The two-size finite-N ladder (Section 6.1) is *consistent with* the findable coefficient recovering toward the existence value $C_3(0) = 0.667$ as N grows, but within grid resolution; the exact $C_K(\rho)$ and the full 3.3× small-β ρ-suppression remain for a faster learner and a finer-grid ladder at $N \geq 750$.

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